SSC MTS 2024 question paper with answers

4 November 2024 · Shift 3

77 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 13 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.4

Mr Rao purchased a laptop for ₹45,000. After a few months, as his office provided another laptop, he sold it at a loss of 12% to Mrs Sharma. After one year, she sold it to an engineering student for a profit of 12%. What is the amount the student had to pay?

(A) ₹34,848
(B) ₹45,000
(C) ₹56,448
(D) ₹44,352

Q.5

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 8% per year rate of interest for 3 years.

(A) 640
(B) 600
(C) 560
(D) 540

Q.6

Two trains are running in opposite directions at the same speed. If the length of each train is 150 m and they cross each other in 15 sec, find the speed of each train (in km/hr).

(A) 18
(B) 36
(C) 42
(D) 24

Q.7

400 chickooes were bought at ₹1280 per hundred and were sold at a profit of ₹980. Find the selling price (in ₹) per dozen of chickooes.

(A) 193
(B) 198
(C) 173
(D) 183

Q.8

What is the ratio in which product A, costing ₹70 per 100 grams, should be mixed with product B, costing ₹100 per 100 grams, so that on selling the mixture at ₹1.20 per gram, a profit of 25% is achieved?

(A) 2 : 13
(B) 9 : 2
(C) 2 : 9
(D) 13 : 2

Q.9

The HCF and LCM of two numbers are 11 and 3465, respectively. If one of the two numbers is 693, what is the other number?

(A) 65
(B) 75
(C) 55
(D) 315

Q.10

If Mohan's income is 20% less than Arvind's income, then how much per cent is Arvind's income more than Mohan's income?

(A) 25%
(B) 15%
(C) 30%
(D) 20%

Q.11

The area of a right-angled triangle is 80m². If its perpendicular is equal to 20 m, then find its base.

(A) 3 m
(B) 6 m
(C) 4 m
(D) 8 m

Q.12

A clothing store announces a discount scheme 'Buy one shirt and get 40% off on another shirt of the same price'. What is the profit percentage of the store, given that each shirt has a cost price 60% below the marked price?

(A) 120%
(B) 100%
(C) 60%
(D) 80%

Q.13

Find the simple interest (in ₹) on ₹2500 as a sum borrowed at 6% per year rate of interest for 2 years.

(A) 340
(B) 300
(C) 240
(D) 260

Q.14

If 80 workers took 25 days to finish a task,how many days will it take to finish the same task if 20 more workers join?

(A) 12
(B) 16
(C) 20
(D) 8

Q.16

In an examination, the average score of a student was 71.9. If the student would have got 28 more marks in English, 22 more marks in Mathematics, 16 more marks in Computer Science, 14 more marks in History, 10 less marks in Hindi, and retained the same marks in other subjects, then the average score would have been 80.65. How many subjects were there in the examination?

(A) 8
(B) 7
(C) 6
(D) 9

Q.17

A train travelling at a speed of 36 km/h crosses a pole in 24 sec. What is the length of the train?

(A) 300 m
(B) 240 m
(C) 30 m
(D) 1.5 m

Q.18

The average weight of X, Y and Z is 67 kg. The average weight of X and Y is 64 kg and that of Y and Z is 52 kg. The weight of Y (in kg) is:

(A) 28
(B) 31
(C) 33
(D) 30

Q.20

Ram borrowed ₹20,000 at 6% per annum and ₹25,000 at 8% per annum, for the same duration. He had to pay ₹9,600 in all as interest. What is the time period?

(A) 4 years
(B) 3 years
(C) 2 years
(D) 5 years

Reasoning Ability and Problem Solving 15 questions

Q.1

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All irons are heaters. Some irons are fans. Conclusions: (I) Some heaters are fans. (II) All fans are irons.

(A) Both conclusion I and II follow
(B) Only conclusion II follows
(C) Neither conclusion I nor II follows
(D) Only conclusion I follows

Q.2

Six persons – Rahul, Ram, Ranjan, Reena, Rohit and Rupali – are sitting around a circular table facing the centre, but not necessarily in the same order. Rupali is sitting second to the right of Ram, who is Reena's immediate neighbour. Rohit is not an immediate neighbour of Reena. Rahul is an immediate neighbour of both, Rupali and Ram. Who is sitting second to the left of Ram?

(A) Ranjan
(B) Rupali
(C) Rahul
(D) Reena

Q.4

18 is related to 91 following a certain logic. Following the same logic, 27 is related to 136. To which of the following is 45 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 - Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 226
(B) 222
(C) 234
(D) 238

Q.5

In a certain code language, 'BIRD' is coded as '6428' and 'DOPE' is coded as '5679'.What is the code for 'D' in the given code language?

(A) 7
(B) 5
(C) 6
(D) 8

Q.6

Select the number from among the given options that can replace the question mark (?) in the following series. 2, 3, 6 ,11, 18 ,?

(A) 31
(B) 30
(C) 27
(D) 29

Q.7

What will come in the place of '?' in the following equation, if '+' and '÷ ' are interchanged and '×' and '–' are interchanged? 88 ÷ 18 × 28 + 4 – 14 =?

(A) 8
(B) –2
(C) 18
(D) 28

Q.8

EBHD is related to IFLH in a certain way based on the English alphabetical order. In the same way, JGMI is related to NKQM. To which of the following is OLRN related, following the same logic?

(A) PSRV
(B) SPRV
(C) PSVR
(D) SPVR

Q.9

In a certain code language, ‘zebra tool bean’ is coded as 'rj pi ki' and ‘bean knife army’ is coded as ‘ko jm pi’. How is ‘bean’ coded in the given language?

(A) rj
(B) ko
(C) pi
(D) ki

Q.10

In a certain code language, 'ABROAD' is coded as ‘WZLIYZ’ and 'ADVISE' is coded as ‘VHREWZ’. How will 'AGENDA' be coded in the same language?

(A) ZWNVUZ
(B) ZTWMWY
(C) ZTVMWZ
(D) ZWMVTZ

Q.11

In the following number-pairs, the second number is obtained by applying certain mathematical operations to the first number. Select the set in which the numbers are related in the same way as are the numbers of the following sets. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding/ subtracting /multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 246, 123; 448, 224

(A) 624, 312
(B) 102, 226
(C) 444, 66
(D) 325, 96

Q.12

What will come in the place of the question mark (?) in the following equation, if ‘+’ and ‘–’ are interchanged and ‘×’ and ‘÷’ are interchanged? $2 \div 40 \times 5 - 4 + 5 = ?$

(A) 11
(B) 13
(C) 31
(D) 15

Q.13

Six people A, B, C, D, E and F are sitting around a circular table, facing the centre (but not necessarily in the same order). F is sitting third to the right of C. D is sitting second to the left of F. A and E are immediate neighbours. E is sitting second to the right of B. How many people are sitting between E and C when counted from the left of E?

(A) 1
(B) 4
(C) 2
(D) 3

Q.14

UD 1 is related to OF –6 in a certain way. In the same way, IX 6 is related to CZ –1. To which of the following is MJ 3 related, following the same logic?

(A) GN –4
(B) GK 4
(C) GM 4
(D) GL –4

Q.15

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: Some tables are plates. Some tables are spoons. Conclusions: (I) At least some spoons are plates. (II) Some plates are tables.

(A) Both conclusions (I) and (II) follow.
(B) Neither conclusion (I) nor (II) follows.
(C) Only conclusion (I) follows.
(D) Only conclusion (II) follows.

Q.16

What should come in place of the question mark (?) in the given series based on the English alphabetical order? HXS, JVQ, LTO, NRM, ?

(A) PPK
(B) PNQ
(C) OPD
(D) PPM

General Awareness 25 questions

Q.1

Who is the author of 'The story of My Experiments with Truth'?

(A) BR Ambedkar
(B) MK Gandhi
(C) Jawaharlal Nehru
(D) Paramahansa Yogananda

Q.2

At 10×10^4 Pa pressure and 300 K temperature, the volume of 0.09 mol CO_2 gas is 22.4×10^{-3} m^3. What will be the volume of the gas if the pressure is decreased to 2×10^4 Pa?

(A) 28.1×10^{-3} m^3
(B) 56.3×10^{-3} m^3
(C) 112.0×10^{-3} m^3
(D) 37.4×10^{-3} m^3

Q.3

Which of the following Articles entails "Equality of opportunity in matters of public employment"?

(A) Article 16
(B) Article 14
(C) Article 17
(D) Article 15

Q.4

To control the prices, which of the following Sultans of Delhi set up three different markets for different commodities in Delhi?

(A) Shamsuddin Iltutmish
(B) Alauddin Khalji
(C) Muhammad bin Tughlaq
(D) Ghiyasuddin Balban

Q.5

The modern-day _____ was molded by the renowned Tanjore Bandhu, consisting of the four Nattuvanars: Ponnayya, Vadivelu, Sivanandam and Chinnayya.

(A) Mohiniyattam
(B) Bharatanatyam
(C) Kathakali
(D) Kuchipudi

Q.6

Which state in India has the largest scheduled tribe population as per the 2011 Census?

(A) Chhattisgarh
(B) Odisha
(C) Jharkhand
(D) Madhya Pradesh

Q.7
The partition of _____was annulled in 1911.

(A) Burma
(B) Punjab
(C) Bengal
(D) United Provinces

Q.8
Who became the first female armless archer to win a medal at the Para World Championship event, 2023?

(A) Shikha Devi
(B) Shreya Devi
(C) Sheetal Devi
(D) Seema Devi

Q.9
Which of the following options represents the correct flow of energy in an ecosystem?

(A) Sunlight → Producers → Herbivores → Carnivores
(B) Sunlight → Herbivores → Carnivores → Producers
(C) Sunlight → Carnivores → Producers → Herbivores
(D) Sunlight → Herbivores → Producers → Carnivore

Q.10
In 1990, who established Nrityagram, a dance school in Bengaluru?

(A) Protima Gauri Bedi
(B) Mukti Mohan
(C) Madhavi Mudgal
(D) Rekha Raju

Q.11
Sawai Gandharva was associated with which of the following gharanas?

(A) Benaras gharana
(B) Agra gharana
(C) Indore gharana
(D) Kirana gharana

Q.12
The percentage of pore space between the particles inside a rock or soil is called:

(A) Porosity
(B) Salinity
(C) Density
(D) Permeability

Q.13
Losar festival is celebrated in which of the following states?

(A) Arunachal Pradesh
(B) Goa
(C) Chhattisgarh
(D) Andhra Pradesh

Q.14
In which state is the Rewa Ultra Mega Solar Power Project located?

(A) Madhya Pradesh
(B) Rajasthan
(C) Karnataka
(D) Tamil Nadu

Q.15
What is the chemical name for the compound responsible for the cooling sensation in mint leaves?

(A) Capsaicin
(B) Caffeine
(C) Ethanol
(D) Menthol

Q.16
In March 2023, who was appointed as Chairperson of Pension Fund Regulatory Development Authority (PFRDA)?

(A) Suresh N Patel
(B) Subodh Jaiswal
(C) Tushar Mehta
(D) Deepak Mohanty

Q.17
In the State Legislature, _____ is subject to dissolution.

(A) House of People
(B) Legislative Council
(C) Council of States
(D) Legislative Assembly

Q.18
For which of the following minerals are Jharia and Raniganj known?

(A) Coal
(B) Copper
(C) Gold
(D) Marble

Q.19
Who among the following formed an organisation to support widow remarriage in Madras Presidency?

(A) Veereshlingam Pantulu
(B) V Chidambaram Pillai
(C) C Rajagopalachari
(D) Ishwar Chandra Vidyasagar

Q.20
The _____ scheme, that covers artisans related to 18 fields, was included under the 'Payments Infrastructure Development Fund' ('PIDF') scheme by the Reserve Bank of India in October 2023.

(A) PM USHA
(B) PM Swanidhi
(C) PM Vishwakarma
(D) PM Ullas

Q.21
What was the name of the official torch for the Khelo India University Games 2022?

(A) Ujala
(B) Mitra
(C) Samarth
(D) Shakti

Q.22
As per Census 2011, which of the following is the most populated city in India?

(A) Kolkata
(B) Chennai
(C) Mumbai
(D) Delhi

Q.23
The second largest dome in the world was built in India and is still standing strong. Which dome is being spoken about here?

(A) Teela
(B) Gol Gumbaz
(C) GlampEco
(D) Calico Dome

Q.24
Which of the following statements is correct?

(A) At burials in Harappan sites the dead were generally laid in pits.
(B) Harappan culture spread only to Sindh and Punjab.
(C) The script of the Harappan culture has been deciphered by historians.
(D) Granaries in the Indus Valley Civilization were found in Harappa only.

Q.25
Identify the correct match in the context of microfinance.

(A) Muhammad Yunus - Grameen Bank - Pakistan
(B) Muhammad Yasir - Grameen Bank - Pakistan
(C) Muhammad Yunus - Grameen Bank - Bangladesh
(D) Muhammad Yasir - Grameen Bank - Bangladesh

English Language and Comprehension 22 questions

Q.1

Select the most appropriate option to fill in the blank. Industrialisation gives even the least talented person a chance to _____ his livelihood.

(A) Break
(B) Earn
(C) Live
(D) Give

Q.2

Select the correctly spelt word to fill in the blank. We _____ to follow the traffic rules.

(A) ought
(B) aught
(C) aut
(D) oght

Q.4

Select the most appropriate ANTONYM of the given word. Obsequious

(A) Ordained
(B) Obstinate
(C) Ostracised
(D) Obliterate

Q.6

Select the option that will improve the underlined part of the given sentence. According to Greek mythology, Sipylus is “holier” city of Lydia.

(A) most holiest
(B) the holiest
(C) the holier
(D) most holy

Q.7

Select the most appropriate meaning of the underlined word. The “sink” is clogged due to solid materials.

(A) To drown in water
(B) A container which holds water
(C) A large fixed container to wash hands or utensils
(D) A pipeline for water supply

Q.8

Select the INCORRECTLY spelt word.

(A) Awful
(B) Barren
(C) Awsome
(D) Bitter

Q.9

Select the most appropriate superlative adjective form to fill in the blanks. What’s ___ book you’ve ever read?

(A) the interesting
(B) more interesting
(C) the most interesting
(D) interesting

Q.10

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. A good lawyer / always looks in the / case before he / attends court.

(A) A good lawyer
(B) always looks in the
(C) attends court
(D) case before he

Q.11

Select the most appropriate meaning of the given idiom. On the back foot

(A) In a hurry
(B) At a disadvantage
(C) Situated farther away
(D) In a relaxed state

Q.12

Select the most appropriate synonym of the given word. Gloomy

(A) Dismal
(B) Buoyant
(C) Sticky
(D) Vain

Q.14

Parts of the following sentence have been given as options. Select the option that contains an error. Before the internet was available on mobile, I read often books for entertainment and information.

(A) was available on mobile
(B) for entertainment and information.
(C) I read often books
(D) Before the internet

Q.15

Select the most appropriate present tense form of the verb to fill in the blank. _____ they like sweets?

(A) Do
(B) Has
(C) Does
(D) Had

Q.16

Select the most appropriate ANTONYM of the given word. Predicament

(A) Redeem
(B) Mess
(C) Docile
(D) Solution

Q.17

The following sentence has been split into four segments. Identify the segment that contains a grammatical/spelling error. They decided to /start their journey / towards the hills / in midnight.

(A) start their journey
(B) They decided to
(C) in midnight
(D) towards the hills

Q.18

Select the most appropriate option to fill in the blank. Despite his initial _____, he eventually agreed to join the team.

(A) reluctance
(B) compassion
(C) determination
(D) enthusiasm

Q.19

Select the most appropriate option to fill in the blank. The investigating team smelled a _____ in the suspect's defence.

(A) dog
(B) cat
(C) rat
(D) bat

Q.20

Select the correctly spelt word to fill in the blank meaningfully.. The _____ on the wall was beautifully painted.

(A) pickchure
(B) pictur
(C) picter
(D) picture

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Opera refers (1)_____ a dramatic art form, originating in Europe, in which the emotional content (2)_____ conveyed to the audience as much through music, both vocal and instrumental, as it is through the lyrics. (3)_____contrast, in musical theatre an actor's dramatic performance is primary, and the music (4)_____ a lesser role. The drama in opera is (5)_____ using the primary elements of theatre such as scenery, costumes and acting.

Select the most appropriate option to fill in blank no. 1.

(A) to
(B) for
(C) on
(D) with

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Opera refers (1)_____ a dramatic art form, originating in Europe, in which the emotional content (2)_____ conveyed to the audience as much through music, both vocal and instrumental, as it is through the lyrics. (3)_____contrast, in musical theatre an actor's dramatic performance is primary, and the music (4)_____ a lesser role. The drama in opera is (5)_____ using the primary elements of theatre such as scenery, costumes and acting.

Select the most appropriate option to fill in blank no. 2.

(A) are
(B) were
(C) is
(D) are being

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Opera refers (1)_____ a dramatic art form, originating in Europe, in which the emotional content (2)_____ conveyed to the audience as much through music, both vocal and instrumental, as it is through the lyrics. (3)_____contrast, in musical theatre an actor's dramatic performance is primary, and the music (4)_____ a lesser role. The drama in opera is (5)_____ using the primary elements of theatre such as scenery, costumes and acting.

Select the most appropriate option to fill in blank no. 3.

(A) Beside
(B) Under
(C) For
(D) In

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Opera refers (1)_____ a dramatic art form, originating in Europe, in which the emotional content (2)_____ conveyed to the audience as much through music, both vocal and instrumental, as it is through the lyrics. (3)_____contrast, in musical theatre an actor's dramatic performance is primary, and the music (4)_____ a lesser role. The drama in opera is (5)_____ using the primary elements of theatre such as scenery, costumes and acting.

Select the most appropriate option to fill in blank no. 4.

(A) play
(B) plays
(C) playing
(D) played

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. Opera refers (1)_____ a dramatic art form, originating in Europe, in which the emotional content (2)_____ conveyed to the audience as much through music, both vocal and instrumental, as it is through the lyrics. (3)_____contrast, in musical theatre an actor's dramatic performance is primary, and the music (4)_____ a lesser role. The drama in opera is (5)_____ using the primary elements of theatre such as scenery, costumes and acting.

Select the most appropriate option to fill in blank no. 5.

(A) presenting

(B) being present

(C) presented

(D) present

Answer key

Numerical and Mathematical Ability

Q.4 **(D)**	Q.5 **(B)**	Q.6 **(B)**	Q.7 **(D)**	Q.8 **(A)**	Q.9 **(C)**
Q.10 **(A)**	Q.11 **(D)**	Q.12 **(B)**	Q.13 **(B)**	Q.14 **(C)**	Q.16 **(A)**
Q.17 **(B)**	Q.18 **(B)**	Q.20 **(B)**			

Reasoning Ability and Problem Solving

Q.1 **(D)**	Q.2 **(A)**	Q.4 **(A)**	Q.5 **(C)**	Q.6 **(C)**	Q.7 **(A)**
Q.8 **(D)**	Q.9 **(C)**	Q.10 **(D)**	Q.11 **(A)**	Q.12 **(D)**	Q.13 **(D)**
Q.14 **(D)**	Q.15 **(D)**	Q.16 **(A)**			

General Awareness

Q.1 **(B)**	Q.2 **(C)**	Q.3 **(A)**	Q.4 **(B)**	Q.5 **(B)**	Q.6 **(D)**
Q.7 **(C)**	Q.8 **(C)**	Q.9 **(A)**	Q.10 **(A)**	Q.11 **(D)**	Q.12 **(A)**
Q.13 **(A)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(D)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(A)**	Q.20 **(C)**	Q.21 **(D)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(A)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(B)**	Q.2 **(A)**	Q.4 **(B)**	Q.6 **(B)**	Q.7 **(C)**	Q.8 **(C)**
Q.9 **(C)**	Q.10 **(B)**	Q.11 **(B)**	Q.12 **(A)**	Q.14 **(C)**	Q.15 **(A)**
Q.16 **(D)**	Q.17 **(C)**	Q.18 **(A)**	Q.19 **(C)**	Q.20 **(D)**	Q.21 **(A)**
Q.22 **(C)**	Q.23 **(D)**	Q.24 **(B)**	Q.25 **(C)**		

Explanations

Numerical and Mathematical Ability

Q.4 · Answer (D) ₹44,352

After the 12% loss: 45,000 × 0.88 = ₹39,600. After the 12% profit: 39,600 × 1.12 = ₹44,352.

Q.5 · Answer (B) 600

SI = 2500 × 8 × 3/100 = ₹600.

Q.6 · Answer (B) 36

Running towards each other, the trains cover 150 + 150 = 300 m in 15 s, a combined 20 m/s. So each runs at 10 m/s = 10 × 18/5 = 36 km/h.

Q.7 · Answer (D) 183

Cost = 4 × ₹1,280 = ₹5,120, so the 400 sell for 5,120 + 980 = ₹6,100, or ₹15.25 each. A dozen: 12 × 15.25 = ₹183.

Q.8 · Answer (A) 2 : 13

Selling at ₹1.20 a gram (₹120 per 100 g) for a 25% profit means a cost of 120/1.25 = ₹96 per 100 g. By alligation A : B = (100 – 96) : (96 – 70) = 4 : 26 = 2 : 13.

Q.9 · Answer (C) 55

The product of two numbers equals HCF × LCM: 11 × 3465/693 = 38,115/693 = 55.

Q.10 · Answer (A) 25%

If Arvind earns 100, Mohan earns 80. Arvind's 20 more is 20/80 = 25% of Mohan's income.

Q.11 · Answer (D) 8 m

Area = ½ × base × perpendicular: 80 = ½ × b × 20, so b = 8 m.

Q.12 · Answer (B) 100%

Take the marked price as ₹100, so each shirt costs ₹40. Two shirts sell for 100 + 60 = ₹160 against a cost of ₹80: a profit of 80/80 = 100%.

Q.13 · Answer (B) 300

SI = 2500 × 6 × 2/100 = ₹300.

Q.14 · Answer (C) 20

The task is 80 × 25 = 2,000 worker-days; 100 workers finish it in 2,000/100 = 20 days.

Q.16 · Answer (A) 8

The changes add 28 + 22 + 16 + 14 – 10 = 70 marks and raise the average by 80.65 – 71.9 = 8.75, so there are 70/8.75 = 8 subjects.

Q.17 · Answer (B) 240 m

36 km/h = 10 m/s. Crossing a pole, the train covers its own length: 10 × 24 = 240 m.

Q.18 · Answer (B) 31

X + Y + Z = 3 × 67 = 201, X + Y = 2 × 64 = 128 and Y + Z = 2 × 52 = 104. Y = 128 + 104 – 201 = 31 kg.

Q.20 · Answer (B) 3 years

One year's interest = 20,000 × 6/100 + 25,000 × 8/100 = 1,200 + 2,000 = ₹3,200, so ₹9,600 takes 9,600/3,200 = 3 years.

Reasoning Ability and Problem Solving

Q.1 · Answer (D) Only conclusion I follows

The irons that are fans are also heaters, so some heaters are fans (I). Nothing says every fan is an iron, so II does not follow.

Q.2 · Answer (A) Ranjan

Facing the centre, each person's right is anticlockwise. Rahul sits between Ram and Rupali, and Reena on Ram's other side. Rohit may not sit next to Reena, so he sits opposite Ram and Ranjan sits next to Reena: second to the left of Ram.

Q.4 · Answer (A) 226

The second number is five times the first plus 1: 18 × 5 + 1 = 91 and 27 × 5 + 1 = 136. Likewise 45 × 5 + 1 = 226.

Q.5 · Answer (C) 6

D is the only letter in both words and 6 the only digit in both codes, so D = 6.

Q.6 · Answer (C) 27

The differences are 1, 3, 5, 7, so the next is 9: 18 + 9 = 27.

Q.7 · Answer (A) 8

After interchanging the signs: 88 + 18 – 28 ÷ 4 × 14 = 88 + 18 – 98 = 8.

Q.8 · Answer (D) SPVR

Each letter moves 4 places forward: E → I, B → F, H → L, D → H, and the same for JGMI → NKQM. So OLRN → SPVR.

Q.9 · Answer (C) pi

'bean' is the only word in both sentences and 'pi' the only code in both, so bean = pi.

Q.10 · Answer (D) ZWMVTZ

Each letter is replaced by its opposite in the alphabet (A ↔ Z, B ↔ Y, and so on) and the result is written backwards: ABROAD → ZYILZW → WZLIYZ. So AGENDA → ZTVMWZ → ZWMVTZ.

Q.11 · Answer (A) 624, 312

The second number is half the first: 246 ÷ 2 = 123 and 448 ÷ 2 = 224. Likewise 624 ÷ 2 = 312.

Q.12 · Answer (D) 15

After interchanging the signs: 2 × 40 ÷ 5 + 4 – 5 = 16 + 4 – 5 = 15.

Q.13 · Answer (D) 3

Facing the centre, each person's right is anticlockwise. F sits opposite C and D second to F's left; E second to the right of B then puts B, E and A in the other seats, so going right from C the order is C, D, B, F, E, A. Counting leftwards from E: F, B and D come before C, three people.

Q.14 · Answer (D) GL –4

The first letter moves 6 back, the second 2 forward and the number falls by 7: U → O, D → F, 1 → –6 and I → C, X → Z, 6 → –1. So MJ 3 → GL –4.

Q.15 · Answer (D) Only conclusion (II) follows.

Some tables are plates, so some plates are tables (II). The tables that are plates and those that are spoons need not overlap, so I does not follow.

Q.16 · Answer (A) PPK

The first letters move 2 forward (H, J, L, N, P), the second 2 back (X, V, T, R, P) and the third 2 back (S, Q, O, M, K): PPK.

General Awareness

Q.1 · Answer (B) MK Gandhi

'The Story of My Experiments with Truth' is MK Gandhi's autobiography.

Q.2 · Answer (C) 112.0×10^{-3} m³

At constant temperature Boyle's law gives $P_1V_1 = P_2V_2$: $V_2 = (10 \times 10^4 \times 22.4 \times 10^{-3})/(2 \times 10^4) = 5 \times 22.4 \times 10^{-3} = 112.0 \times 10^{-3}$ m³. (The amount of gas and the temperature are not needed.)

Q.3 · Answer (A) Article 16
Article 16 guarantees equality of opportunity in matters of public employment. Article 14 is equality before the law, Article 15 bars discrimination and Article 17 abolishes untouchability.

Q.4 · Answer (B) Alauddin Khalji
Alauddin Khalji's market reforms set up separate markets in Delhi for grain, for cloth and other goods, and for horses, slaves and cattle, all at fixed prices.

Q.5 · Answer (B) Bharatanatyam
The Tanjore Quartet, four brothers at the Thanjavur court in the early 19th century, set the order of items of modern Bharatanatyam.

Q.6 · Answer (D) Madhya Pradesh
By the 2011 Census Madhya Pradesh has the largest scheduled tribe population of any state, about 1.53 crore.

Q.7 · Answer (C) Bengal
The partition of Bengal, made in 1905, was annulled in 1911, as announced at the Delhi Durbar.

Q.8 · Answer (C) Sheetal Devi
Sheetal Devi won silver in the women's compound event at the 2023 World Archery Para Championships in Pilsen, the first female armless archer to win a medal there.

Q.9 · Answer (A) Sunlight → Producers → Herbivores → Carnivores
Energy enters an ecosystem through producers, which capture sunlight, and passes to herbivores and then to carnivores.

Q.10 · Answer (A) Protima Gauri Bedi
Odissi dancer Protima Gauri Bedi set up Nrityagram, a residential dance village near Bengaluru, in 1990.

Q.11 · Answer (D) Kirana gharana
Sawai Gandharva (Rambhau Kundgolkar), a disciple of Abdul Karim Khan, was a singer of the Kirana gharana and the guru of Bhimsen Joshi.

Q.12 · Answer (A) Porosity
Porosity is the share of a rock's or soil's volume taken up by pore spaces; permeability is how easily water passes through it.

Q.13 · Answer (A) Arunachal Pradesh
Losar, the Tibetan Buddhist new year, is celebrated in Arunachal Pradesh by communities such as the Monpas, as well as in Sikkim and Ladakh.

Q.14 · Answer (A) Madhya Pradesh
The 750 MW Rewa Ultra Mega Solar project is in Rewa district, Madhya Pradesh.

Q.15 · Answer (D) Menthol
Menthol gives mint its cooling sensation. Capsaicin is what makes chillies hot.

Q.16 · Answer (D) Deepak Mohanty
Deepak Mohanty, a former executive director of the RBI, was appointed Chairperson of the PFRDA in March 2023.

Q.17 · Answer (D) Legislative Assembly
A state's Legislative Assembly can be dissolved; its Legislative Council, like the Rajya Sabha, is a permanent house. The House of the People and the Council of States are the houses of Parliament.

Q.18 · Answer (A) Coal
Jharia (Jharkhand) and Raniganj (West Bengal) are major coalfields of the Damodar valley.

Q.19 · Answer (A) Veereshlingam Pantulu
In the Telugu-speaking areas of the Madras Presidency, Kandukuri Veeresalingam Pantulu formed an association for widow remarriage. Vidyasagar led the movement in Bengal.

Q.20 · Answer (C) PM Vishwakarma
In October 2023 the RBI extended its Payments Infrastructure Development Fund scheme to the beneficiaries of PM Vishwakarma, the scheme for artisans and craftspeople in 18 trades.

Q.21 · Answer (D) Shakti
SSC's key gives 'Shakti' as the name of the official torch of the Khelo India University Games 2022, which Uttar Pradesh hosted in May-June 2023.

Q.22 · Answer (C) Mumbai
By the 2011 Census Greater Mumbai, with about 1.24 crore people, was India's most populous city.

Q.23 · Answer (B) Gol Gumbaz
Gol Gumbaz, the tomb of Mohammed Adil Shah at Bijapur (Vijayapura), Karnataka, is often described as the second largest dome in the world.

Q.24 · Answer (A) At burials in Harappan sites the dead were generally laid in pits.
At Harappan burials the dead were generally laid in pits. The culture also spread to Gujarat, Rajasthan and Haryana, its script is still undeciphered, and a granary was found at Mohenjodaro too.

Q.25 · Answer (C) Muhammad Yunus - Grameen Bank - Bangladesh
Muhammad Yunus founded the Grameen Bank in Bangladesh, a pioneer of microcredit; he and the bank shared the 2006 Nobel Peace Prize.

English Language and Comprehension

Q.1 · Answer (B) Earn
One 'earns' a livelihood.

Q.2 · Answer (A) ought
'Ought to' expresses duty: 'We ought to follow the traffic rules.' 'Aught' means anything.

Q.4 · Answer (B) Obstinate
'Obsequious' means too eager to please or obey; of these, 'obstinate', stubbornly refusing to give way, is the nearest to its opposite.

Q.6 · Answer (B) the holiest
Picking out one city from all the cities of Lydia calls for the superlative with 'the': 'Sipylus is the holiest city of Lydia.' 'Most holiest' is a double superlative.

Q.7 · Answer (C) A large fixed container to wash hands or utensils
A 'sink' is a fixed basin for washing hands or utensils.

Q.8 · Answer (C) Awsome
The correct spelling is 'awesome'.

Q.9 · Answer (C) the most interesting
The superlative of 'interesting' is 'the most interesting'.

Q.10 · Answer (B) always looks in the
To examine a matter closely is to 'look into' it: 'always looks into the case'.

Q.11 · Answer (B) At a disadvantage
To be 'on the back foot' is to be at a disadvantage, forced to defend oneself.

Q.12 · Answer (A) Dismal
'Gloomy' means dark or depressing: 'dismal'. 'Buoyant' is close to its opposite.

Q.14 · Answer (C) I read often books
An adverb of frequency goes before the main verb: 'I often read books'.

Q.15 · Answer (A) Do
With 'they', a present tense question takes 'do': 'Do they like sweets?'

Q.16 · Answer (D) Solution
A 'predicament' is a difficult situation; of these, 'solution', the way out of one, is the nearest to its opposite. 'Mess' is close in meaning.

Q.17 · Answer (C) in midnight
A point of time takes 'at': 'at midnight', not 'in midnight'.

Q.18 · Answer (A) reluctance
'Despite' signals a contrast: his initial 'reluctance', then his agreement.

Q.19 · Answer (C) rat
To 'smell a rat' is to suspect that something is wrong.

Q.20 · Answer (D) picture
The correct spelling is 'picture'.

Q.21 · Answer (A) to
The verb 'refer' takes 'to': 'Opera refers to a dramatic art form'.

Q.22 · Answer (C) is
'The emotional content' is singular, so the passive is 'is conveyed', matching 'as it is through the lyrics'.

Q.23 · Answer (D) In
The fixed phrase is 'In contrast'.

Q.24 · Answer (B) plays
'The music' is singular, so 'plays a lesser role'.

Q.25 · Answer (C) presented
The drama 'is presented' (passive) using scenery, costumes and acting.

This paper online, with tap-to-reveal answers: https://parikshasaathi.com/exams/ssc-mts/question-papers/mts-2024-4-nov-2024-shift-3/
Every solved SSC MTS paper: https://parikshasaathi.com/exams/ssc-mts/question-papers/
Daily current affairs PDF, free, every morning on Telegram: t.me/parikshasaathiapp
Questions and official answers: the Staff Selection Commission (SSC). Explanations: drafted with AI assistance by Pariksha Saathi and checked against the official answers. Pariksha Saathi is not affiliated with any exam body. Spotted an error? Write to sriwastava2@gmail.com.